TANISH SANGHVI

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LinkedIn | GitHub | Portfolio

Professional Summary

Final-year Automation and Robotics Engineering student with hands-on experience developing full-stack applications, backend services, REST APIs, database-driven systems and automated data pipelines. Proficient in Python, JavaScript, TypeScript, React, Next.js, Flask, Node.js and SQL/NoSQL databases. Experienced in authentication, API validation, software testing, debugging, AI integration and application deployment. National hackathon finalist with technical coordination experience.

Education

Vivekanand Education Society's Institute of Technology, Mumbai 2023 – 2027 (Expected)
B.E. in Automation and Robotics Engineering CGPA: 7.43/10.0
Relevant Coursework: Data Structures and Algorithms, Database Management Systems, Machine Learning, Generative AI

Technical Skills

Programming Languages: Python, JavaScript (ES6+), TypeScript, C++, SQL
Frontend Development: React.js, Next.js, HTML5, CSS3, Tailwind CSS
Backend Development: Node.js, Express.js, Flask, REST APIs, Pydantic
Databases: PostgreSQL, MongoDB, Firebase Firestore, Supabase
AI Integration: Gemini API, Generative AI Integration, AI-Powered Application Workflows
Testing and Quality Assurance: Pytest, Jest, Playwright, Unit Testing, Integration Testing, End-to-End Testing
Security and Reliability: JWT Authentication, Role-Based Access Control, API Validation, Error Handling, Debugging
Developer Tools and Deployment: Git, GitHub, GitHub Actions, Postman, Vercel, Render

Projects

Job Hunter – Automated Job Discovery Platform Live Demo | GitHub
Python, Flask, PostgreSQL, Supabase, GitHub Actions

- Built a modular job-ingestion pipeline with configurable parsing adapters for 114 company career-page sources across 9 applicant tracking system platforms.
- Integrated PostgreSQL/Supabase data persistence, configurable job filters and AI-assisted profile scoring to support job discovery and evaluation.
- Implemented error handling, exponential retry backoff and automated tests covering data extraction, persistence integrity and network-failure recovery.

Department Ledger Portal – Academic Records System Live Demo | GitHub
Next.js, React, Firebase Firestore, Jest, Playwright

- Developed an academic records platform with separate student, faculty and administrator workflows for record compilation and portfolio management.
- Implemented role-based access control, audit logging and document autofill using parsed data for structured academic record management.
- Wrote 199 Jest unit tests and 27 Playwright end-to-end tests to validate application logic and user workflows.

Edvanta – AI-Assisted Learning Platform Live Demo | GitHub
Python, Flask, React, MongoDB, Firebase, Pydantic

- Designed and deployed 33 modular REST endpoints supporting assessment workflows, AI-assisted assessment generation and personalized learning pathways.
- Integrated Gemini API to support AI-powered educational workflows and personalized learning experiences.
- Implemented Pydantic request validation, Firebase JWT authentication, protected-route middleware and role-based authorization for controlled API access.

DineEase – Restaurant Ordering System Live Demo | GitHub
React, Node.js, Express.js, MongoDB

- Developed a full-stack restaurant ordering application with menu browsing, cart management, checkout and administrator dashboards.
- Implemented role-protected administrator workflows and MongoDB-backed order processing with persistent application data.
- Wrote 35 automated tests covering order calculations, order-payload handling and route protection.

Battery State Estimator – Battery Analytics Platform Live Demo | GitHub
Python, Flask, MongoDB, NumPy, Data Processing, Visualisation

- Developed a software-based battery simulation and telemetry pipeline generating voltage, current and temperature data across configurable drive cycles and fault scenarios.
- Implemented Echo State Network-based SOC/SOH estimation and comparative analysis against Extended Kalman Filter, Coulomb Counting and adaptive RLS methods.
- Integrated simulation outputs and results from 4 estimation methods with MongoDB and a Flask-based visualisation dashboard for monitoring battery behaviour and comparing model performance.

Achievements & Leadership

- **National Finalist – Hack Celestial 2.0:** Reached the national finals with Edvanta in a competition involving 320+ participating teams.
- **Co-organizer – UniMerge 1.0:** Coordinated technical operations, submission validation and developer support for a national hackathon with 130+ participants.